Dynamic Pricing Strategy Simulator

E-commerce Price Elasticity · Scenario Engine · Optimal Pricing · Revenue Optimization Dashboard

Data Source
Kaggle Online Retail II CSV

Scope
500K+ transactions · 8 categories · 3 years

Tools
Python · SciPy · stats models · NumPy · Power BI

Duration
4 weeks · ~4 hrs./ week

Submission
Report · Dashboard · Slides · Presentation

RATIONALE & RESEARCH QUESTIONS

Problem

Pricing is the highest-leverage retail decision — a 1% improvement can lift profit by 8-11%. Most businesses use cost-plus heuristics. This project builds a data-driven simulator that quantifies demand elasticity and tests revenue impact before any price change is applied.

Objective

Estimate price elasticity per category, build regression models, construct a scenario engine for testing 3 price change levels, and deliver a Power BI dashboard with optimal price recommendations and BSC KPIs.

Research Questions

- **RQ1** What is the price elasticity per category — elastic (e<-1) vs inelastic (e>-1)?
- **RQ2** Is the price-demand curve linear, or does log-log regression reveal non-linearity?
- **RQ3** What revenue and margin impact do 5%, 10%, and 15% price changes generate?
- **RQ4** Do seasonal patterns (month/quarter) significantly moderate price-demand?
- **RQ5** Which categories support a price increase vs which benefit from a discount?

Pricing Decision Logic

Elastic (e < -1): Demand sensitive → discount grows revenue

Inelastic (e > -1): Demand stable → price increase lifts margin

Constraints: Min margin 20% + max demand drop 30% enforced

Ethical Standards
Open Kaggle dataset only · No proprietary data · All assumptions documented · Scenarios labelled as simulations

DATA, METHODOLOGY & KPI FRAMEWORK

Data Source

Dataset: Online Retail II — UCI Machine Learning Repository
Link: kaggle.com/datasets/mashlyn/online-retail-ii-uci
Format: CSV · 500K+ rows · Price, Quantity, StockCode, InvoiceDate
Key fields: Price, Quantity, StockCode, Description, InvoiceDate

Balanced Scorecard KPI Framework

KPIs with RAG logic powering the Power BI tile dashboard:

Perspective	KPI	Target	RAG Logic
Financial	Revenue Impact	> 0%	RED decline · AMB flat · GRN uplift
Profitability	Gross Margin %	> 20%	RED <20% · AMB 20-25% · GRN >25%
Customer	Demand Change	< -30%	RED if demand drop >30% post rise
Strategy	Price Effectiveness	Top 3	Category in top 3 margin rank

Methodology — CRISP-DM

1. Business understanding — Define elasticity, KPIs, simulation constraints
2. Data understanding — Audit 500K rows: returns, zero prices, outliers
3. Data preparation — Remove returns, clip outliers, aggregate weekly
4. Modelling — Log-log regression + scenario engine + optimal price calc
5. Evaluation — R-squared, RMSE, sensitivity analysis, margin check
6. Deployment — Power BI: scenario slider + recommendation cards

Analytics Techniques

- Descriptive stats — price/qty distributions per category
- Log-log regression — price elasticity coefficient estimation
- OLS regression — demand curve with seasonality dummies
- Hypothesis test — elasticity coefficient t-test significance
- Scenario simulation — revenue delta at 5/10/15% price change
- RAG logic — actual vs target per BSC KPI · Cannibalization noted

PLAN, DASHBOARD & DELIVERABLES

Dashboard Layout (Power BI)

Revenue Impact vs baseline	**Margin Change** % vs threshold
Best Strategy penet/premium	**Most Elastic Cat.** category name

Row 2: Demand curve visual · Scenario comparison bar chart
Row 3: Category elasticity ranking · Optimal price recommendations table

4-Week Plan

Week 1 Data & KPI Setup	Download Online Retail II CSV · Audit 500K rows for returns and outliers · Define BSC KPIs + RAG thresholds · Aggregate weekly price-qty pairs · Data Dictionary
Week 2 EDA & Elasticity	Price vs qty scatter plots · Log-log OLS (statsmodels) · Elasticity coefficients + t-test · Classify elastic vs inelastic · Seasonality dummies
Week 3 Simulation & Dashboard	Scenario engine: 5/10/15% price change → revenue + margin delta · Constraints enforced · Optimal price per category · Power BI: KPI tiles (RAG) + demand curve + scenario chart
Week 4 Report & Viva	Write 1,500-2,500 word report · 10 slides with Executive Insights · Viva notes · Timed practice run

Executive Insights (Required)

- Category A (inelastic): 10% price rise with <5% demand loss — estimated +$X monthly margin uplift
- Category B (e=-1.8): a 10% discount grows revenue ~18% due to demand surge effect
- Portfolio strategy: raise 3 inelastic + discount 2 elastic categories → net margin uplift of Y%

Deliverables

- **D1** Raw data + source log + KPI definition sheet (Wk 1)
- **D2** Cleaned CSV + Data Dictionary — all fields described (Wk 1)
- **D3** Python notebook — EDA, log-log regression, elasticity table (Wk 2)
- **D4** Scenario engine + optimal price output + sensitivity (Wk 3)
- **D5** Power BI dashboard — live link + PDF with RAG tiles (Wk 3)
- **D6** Report + 10 slides with Executive Insights section (Wk 4)